

Software Engineer

TECHNICAL SKILLS

Programming Languages: Java, Python, C/C++, Go, C#, SQL (PostgreSQL/MySQL), Bash/Shell Scripting

Libraries & Frameworks: PyTorch, scikit-learn, pandas, NumPy, OpenCV, Flask, FastAPI, Protocol Buffers

System & Infrastructure: Apache Kafka, Telegraf, InfluxDB, Grafana, Redfish, systemd

Cloud & DevOps: Git, GitLab CI/CD, Docker/Podman, Ansible, AWS

AI & Data: ChromaDB, RAG, ETL pipelines

Tools: GitHub Copilot, ChatGPT, opencode, Pytest, JUnit, LaTeX, Agile, Jira

EXPERIENCE

High Performance Computing (HPC) Engineering Intern

May 2026 – Present

Los Alamos National Laboratory

Los Alamos, NM

- Designed and deployed a hardware telemetry pipeline using Redfish API, Kafka, InfluxDB, and Telegraf, collecting 830K+ data points daily with zero data loss through containerized microservices and systemd automation.
- Reverse-engineered Intel Redfish API, implementing telemetry polling and event subscriptions across 6 endpoint families with a Python Flask receiver using dual-write pattern (Kafka + file storage).
- Provisioned a multi-node Rocky Linux cluster using OpenCHAMI, automating OS deployment via Kickstart and cloud-init, and implementing a GitLab CI/CD pipeline to automatically build and test boot images on commit.

Extended Reality (XR) Developer

July 2024 – May 2026

Mesh Labs (formerly Meteor Studio)

Mesa, AZ

- Containerized a GPU-enabled multi-dependency environment using Docker Compose, reducing setup time from 60 to 16 minutes and eliminating Linux/WSL2 compatibility issues.
- Developed a Python 3D camera calibration pipeline using RANSAC-based feature matching and ICP point cloud registration for accurate multi-view scene reconstruction.

Undergraduate Researcher

June 2023 – August 2023

Texas A&M-Corpus Christi

Corpus Christi, TX

- Developed multi-drone control system for PX4 drones in Gazebo simulation, debugging timing and communication across hardware/software boundaries while coordinating up to 10 autonomous drones.
- Designed windowing-based preprocessing pipelines for time-series sensor data, trained and validated KNN, SVM, Random Forest, and Naive Bayes ensemble models, improving gesture recognition accuracy by 10%.

PROJECTS

gRPC Microservices System

- Architected a distributed system with 5 gRPC-based microservices, implementing both interactive CLI and automated client interfaces with robust error handling and modular service design.
- Implemented 5 gRPC services (Fitness, RecipeBook, Sort, Joke, Echo) with Protocol Buffers, file-based persistence, a service registry for discovery, and thread-safe concurrent request handling.
- Deployed system on AWS EC2 instance with public accessibility, enabling multi-client access and validating concurrent service requests.

AI-Integrated Wireless Emergency Alerts

- Engineered a FastAPI microservice integrating retrieval-augmented generation (RAG) with ChromaDB to enable semantic search and context-aware LLM responses.
- Designed a containerized automation system with custom scheduling and health endpoints to orchestrate data ingestion, scraping, and LLM pipelines.
- Developed a fault-tolerant Python scraper for the PBS WARN API, utilizing exponential backoff to reliably extract and normalize live emergency data.
- Built a ChromaDB cleanup service that parses expiration timestamps and batch-deletes stale alerts on a 2-hour schedule, with a health endpoint verifying cleanup status.

Gym Calendar Application (Memoranda) Modernization

- Integrated Google Calendar API with OAuth2 to synchronize cross-timezone class schedules, implementing push-based event creation and resolving datetime offset conflicts.
- Refactored legacy Java codebase (25,000+ LOC) to eliminate compiler warnings and improve maintainability while managing Git workflows and merge conflicts.

EDUCATION

Bachelor of Science, Software Engineering

Arizona State University, Polytechnic

May 2026

Mesa, AZ